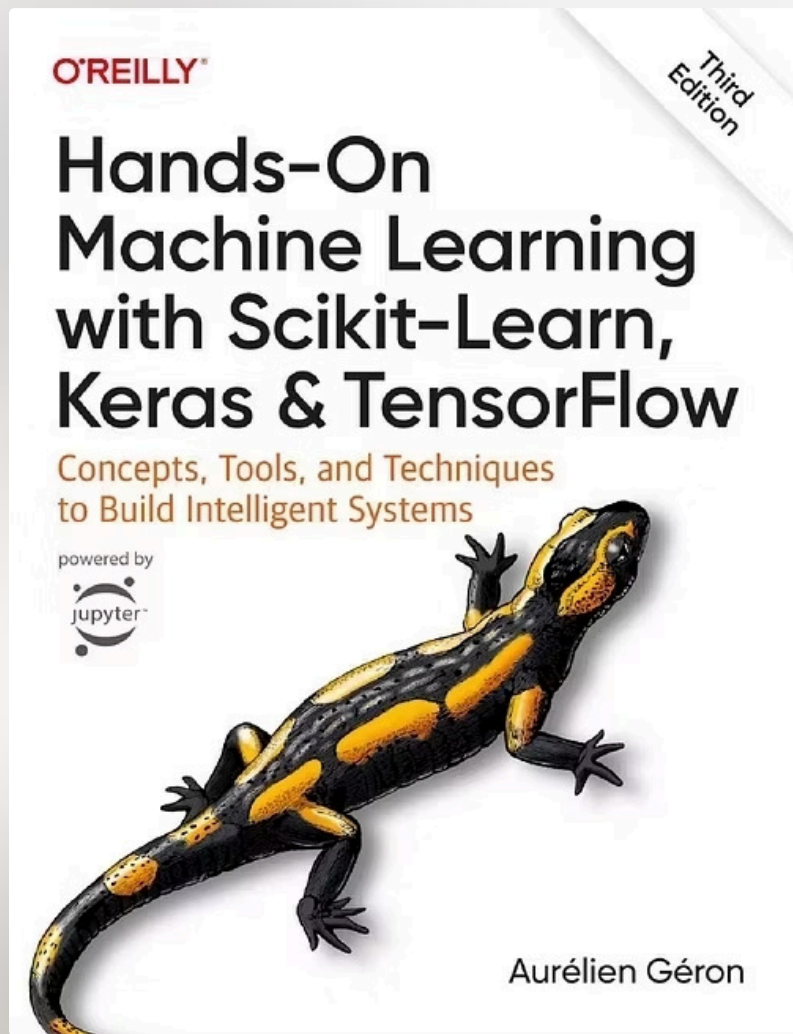




Machine Learning

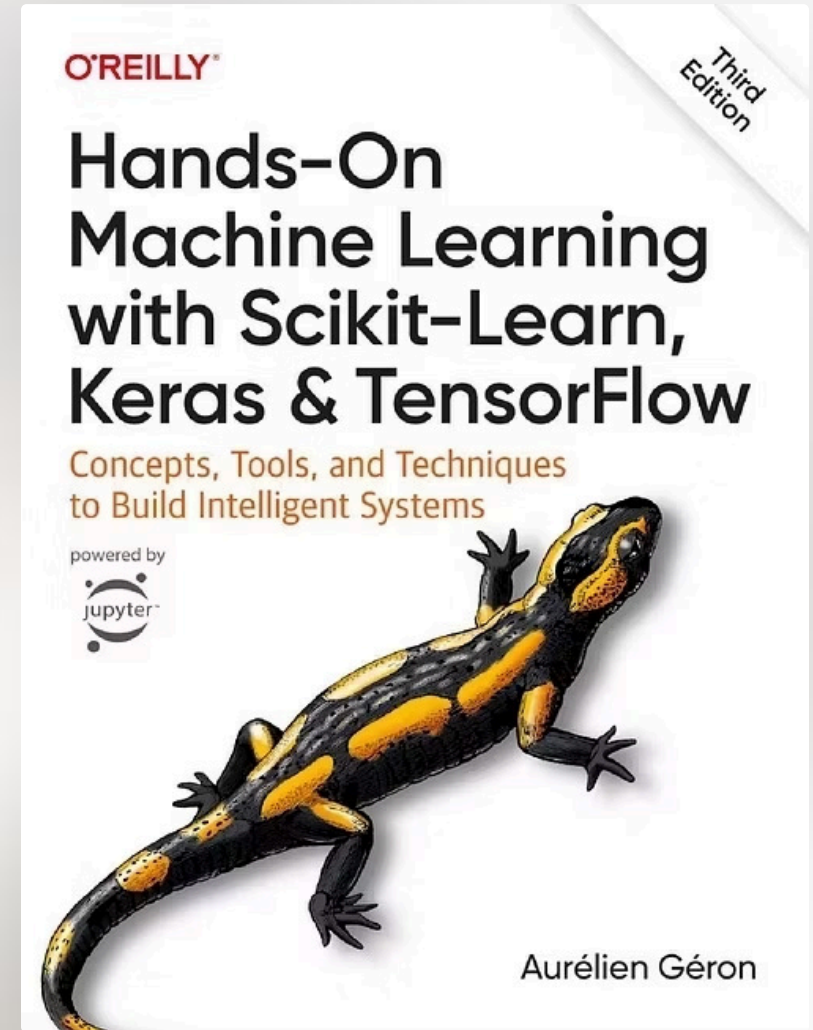
Lessons based on the Textbook "Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow" by Aurélien Géron, 2023.

RP by Rodrigo Martins Pagliares

Last update: 07/22/2025

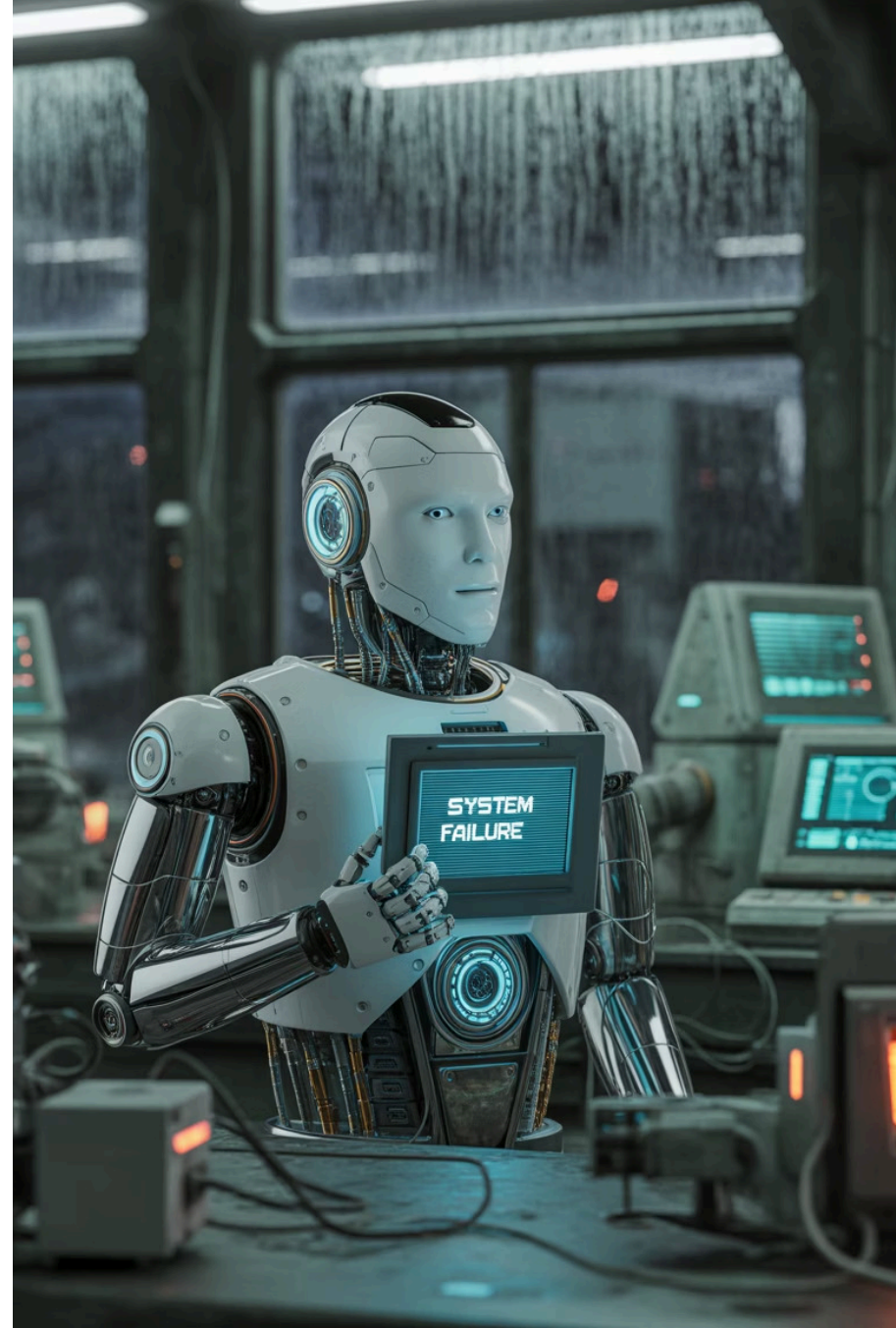
Machine Learning

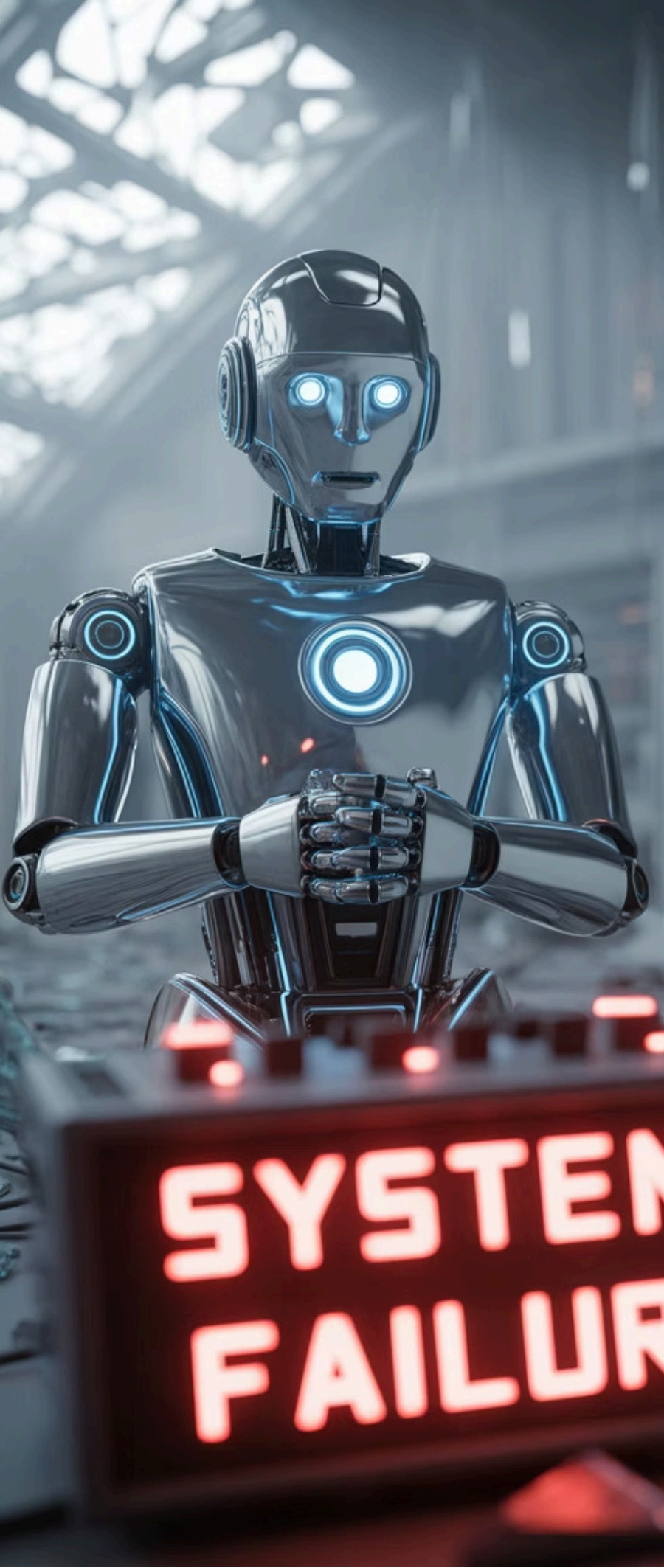
Chapter 1 – The Machine Learning Landscape



Agenda for Chapter 1

1. What Is Machine Learning?
2. Why Use Machine Learning?
3. Examples of Applications
4. Types of Machine Learning Systems
5. **Main Challenges of Machine Learning**
6. Testing and Validating





Agenda for Chapter 1

1. What Is Machine Learning?
2. Why Use Machine Learning?
3. Examples of Applications
4. Types of Machine Learning Systems
 - Classification criteria #1: Supervision during training
 - Supervised, Unsupervised, Semi-Supervised, Self-Supervised, and Reinforcement Learning
 - Classification criteria #2: Approach to Learning Method
 - Batch and Online (Incremental) Learning
 - Classification criteria #3: Learning Method
 - Instance-Based and Model-Based Learning
5. Main Challenges of Machine Learning
 - Insufficient Quantity of Training Data
 - Non-representative Training Data
 - Poor-Quality Data
 - Irrelevant Features
 - Overfitting the Training Data
 - Underfitting the Training Data
 - Stepping Back
6. Testing and Validating
 - Hyperparameter Tuning and Model Selection
 - Data Mismatch

Main Challenges of Machine Learning

Main Challenges of Machine Learning

- Since your main task is to **select a model** and **train it on some data**, the two things that can go wrong are:
 - **Bad Model**
 - **Bad Data**
- **Focus:** Let's start by discussing **bad data**, specifically **insufficient training data**.





Insufficient Quantity of Training Data

- **Problem:**
 - ML algorithms typically require a lot of data to perform well, especially for complex problems.
- **Simple Learning (Human Example):**
 - A toddler learns what an **apple** is after seeing a few examples.
 - ML algorithms require thousands to millions of examples, even for simple tasks.

The Unreasonable Effectiveness of Data

- **Microsoft Research (2001):**
 - **Michele Banko and Eric Brill** showed that **ML algorithms** perform similarly well on complex problems once given enough data (regardless of the algorithm's complexity).
 - **Key takeaway: Data quality and quantity** often matter more than the **algorithm** for solving complex problems.

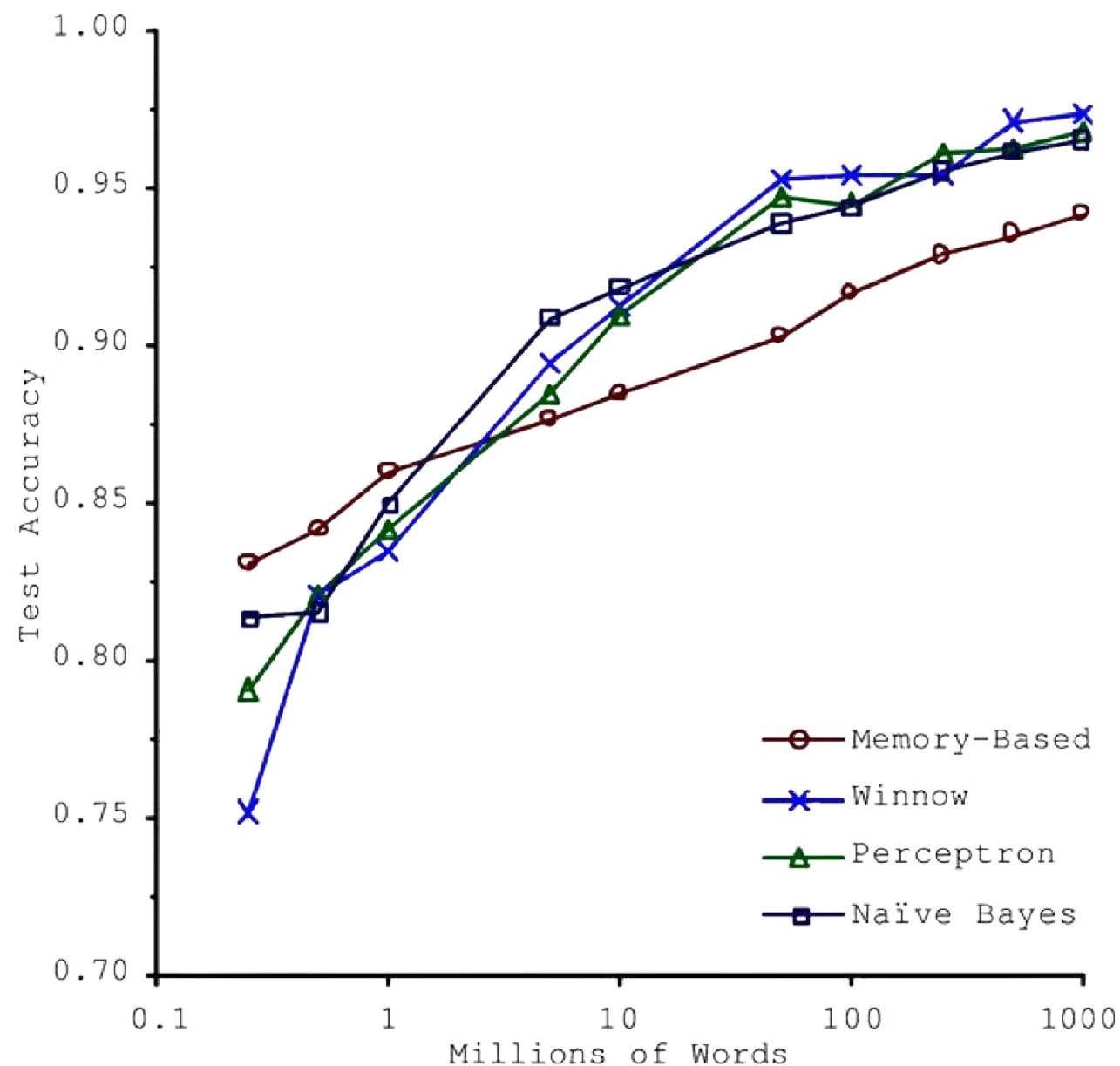
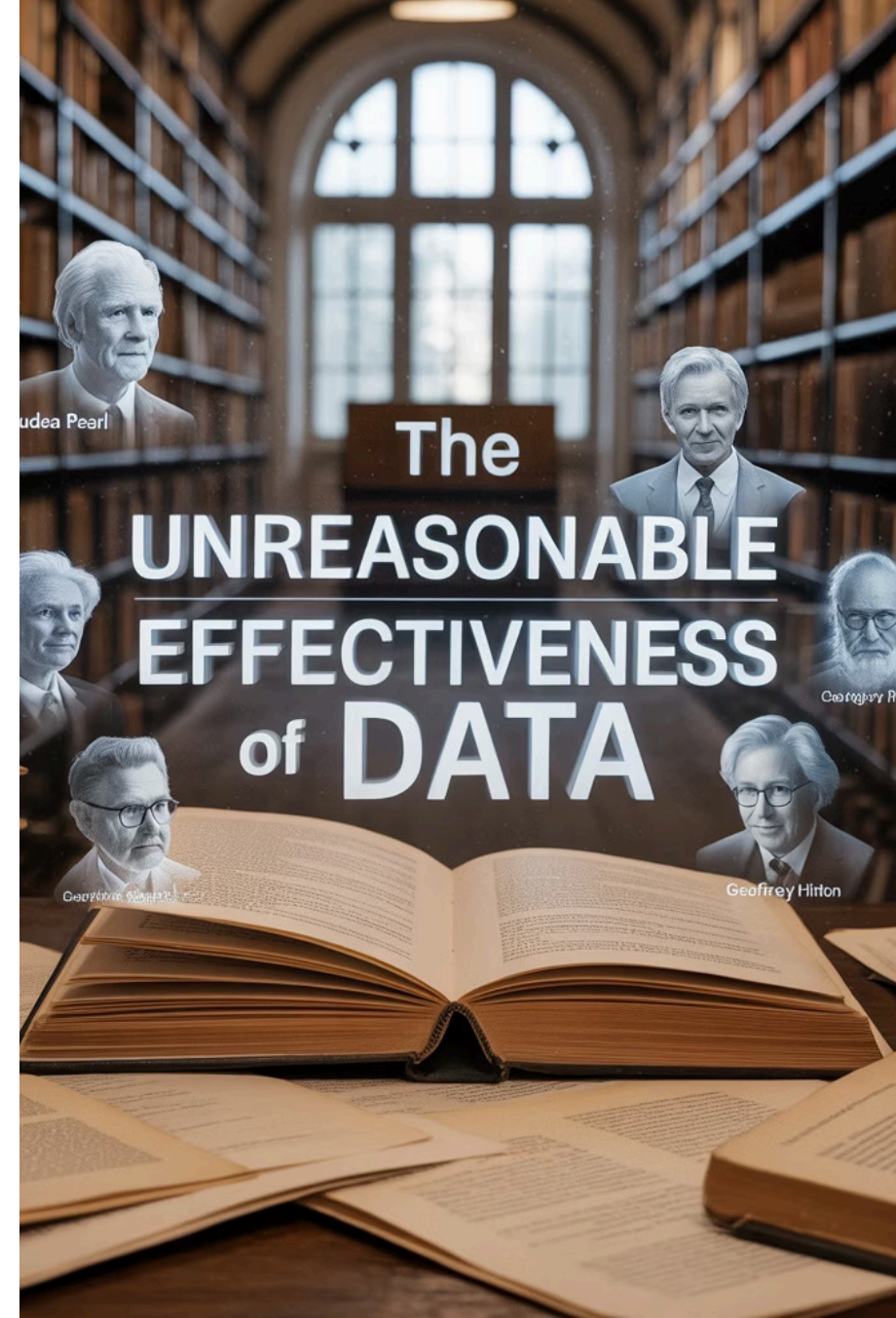
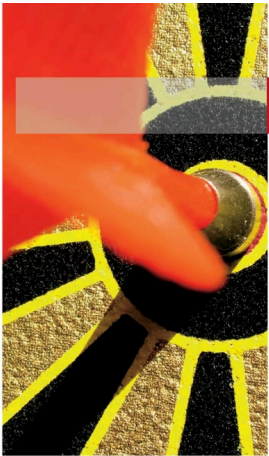


Figure 1-21. The importance of data versus algorithms

The Unreasonable Effectiveness of Data

- Peter Norvig et. al (2009):
 - The idea of the unreasonable effectiveness of data was further emphasized in a paper by Peter Norvig et. al.
 - It shows that more data often leads to better performance than improving the algorithm.





EXPERT OPINION

Contact Editor: **Brian Brannon**, bbrannon@computer.org

The Unreasonable Effectiveness of Data

Alon Halevy, Peter Norvig, and Fernando Pereira, *Google*

Eugene Wigner's article "The Unreasonable Effectiveness of Mathematics in the Natural Sciences"¹ examines why so much of physics can be neatly explained with simple mathematical formulas

such as $f = ma$ or $e = mc^2$. Meanwhile, sciences that involve human beings rather than elementary particles have proven more resistant to elegant mathematics. Economists suffer from physics envy over their inability to neatly model human behavior. An informal, incomplete grammar of the English language runs over 1,700 pages.² Perhaps when it comes to natural language processing and related fields, we're doomed to complex theories that will never have the elegance of physics equations. But

behavior. So, this corpus could serve as the basis of a complete model for certain tasks—if only we knew how to extract the model from the data.

Learning from Text at Web Scale

The biggest successes in natural-language-related machine learning have been statistical speech recognition and statistical machine translation. The reason for these successes is not that these tasks are easier than other tasks; they are in fact much harder than tasks such as document classification that extract just a few bits of information from each document. The reason is that translation is a natural task routinely done every day for a real human need (think of the operations of the European Union or of news agencies). The same is true of speech tran-

Paper: The Unreasonable Effectiveness of Data

2,587 Citations accordingly to Google Scholar (07/22/2025)

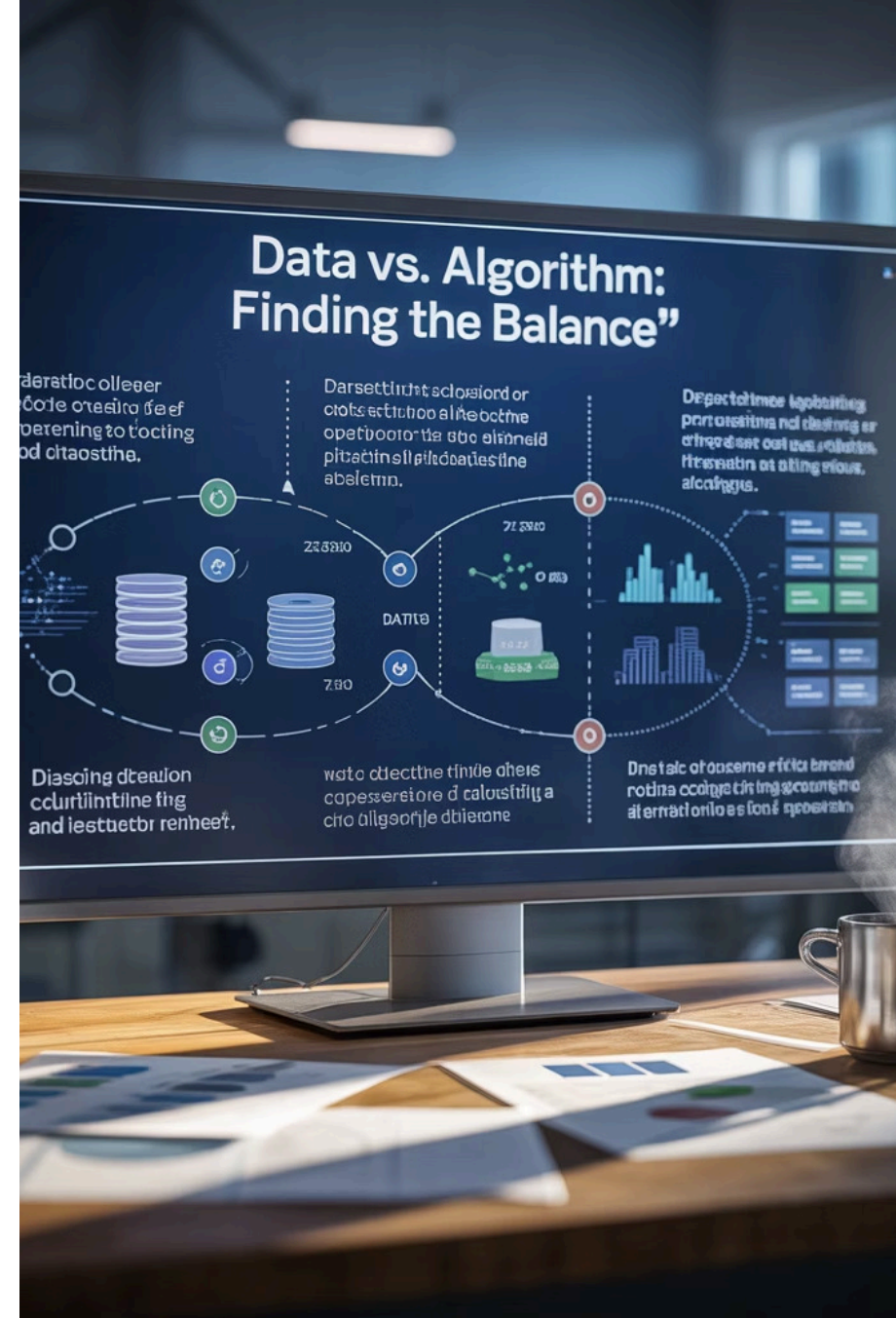
Limitations of Small and Medium-Sized Datasets

- **Challenges:**
 - While **large datasets** are powerful, **small and medium-sized datasets** are still very common.
 - It is often **difficult** and **expensive** to acquire **additional training data**, especially in certain domains.



Key Takeaways

- **Data vs. Algorithm:**
 - For complex problems, the amount of **data** is often more important than the **algorithm** used.
- **Small Datasets:**
 - Despite the power of **large datasets**, **small datasets** are still **prevalent**, and finding more data can be a challenge.
- **Balance Between Algorithm and Data:**
 - Don't abandon improving algorithms just because data is key; the **trade-off** between **data** and **algorithm development** should be considered.





**Next: Non-
representative
Training Data**

Q&A



Open Floor

Open floor for discussion or clarifications



Discussion

Share your thoughts and questions about AI concepts